

From Micro-Failure Tags to Defensive Syndromes Supplementary Operational Templates and Matrices

Companion to: “From Micro-Failure Tags to Defensive Syndromes: A Technical Framework for the Core Six User-Facing Failure Modes in AI Assistants”

Author: Ernesto A. Taylor

ORCID: <https://orcid.org/0009-0003-8766-1070>

Affiliation: YIM Project (Independent Research) - Atlanta, GA

Date: April 5, 2026

DOI: <https://doi.org/10.5281/zenodo.19423182>

Contact: <mailto:research@yeahitsme.com>

Website: <https://yeahitsme.com>

Version: 2.1.6 (Authoritative Long Version)

Word Count: Approx. 31,200

Type: Technical Framework Paper

Classification: Technical Research Framework (Public Release)

Project Timeline: 2024–2026 (continuous AI-assisted development and observation)

Primary Empirical Window: October–December 2025 (n=105 collected; n=45 with complete syndrome coding at publication)

Got it. Here's the reformatted block:

Version History

v2.1.6 - 2026-05-05 Companion version synchronized to main paper v2.1.6 after content correction patch (8 corrections). No changes to operational templates or calibration guidance.

v2.1.5 - 2026-05-02 Companion version synchronized to main paper v2.1.5 after Section 8 heading-style promotion pass. No content changes to operational templates or calibration guidance.

v2.1.4 - 2026-05-02 Companion version synchronized to main paper v2.1.4 after Section 6/7 heading-style promotion pass. No content changes to operational templates or calibration guidance.

v2.1.3 - 2026-05-02 Companion version synchronized to main paper v2.1.3 after the Section 4 micro-failure list compression and package synchronization pass. No content changes to operational templates or calibration guidance.

v2.1.2 - 2026-05-02 Companion version synchronized to main paper v2.1.2 after the Published-folder formatting and package synchronization pass. No content changes to operational templates or calibration guidance.

v2.1.1 - 2026-04-30 Companion version synchronized to main paper v2.1.1 after the consecutive figure-renumbering correction was promoted into a package patch release. No content changes to operational templates or calibration guidance.

v2.1.0 - 2026-04-28 Companion version synchronized to main paper v2.1.0 after the accumulated post-v2.0.12 published-paper revisions were promoted into a new release marker. No content changes to operational templates or calibration guidance.

v2.0.12 - 2026-04-26 Combined mechanical and prose fluidity correction pass. Mechanical: fixed stale micro-failure tag count in s.7.3 (20->44); corrected Supplementary Materials footer version (v2.0.6->v2.0.12); corrected Verification Appendix citation block (v2.0.10->v2.0.12); corrected next-review date to July 26, 2026; fixed punctuation errors in s.2.1, s.2.2, and s.8.2 heading; removed orphaned Figure 1 reference at end of s.1.2. Prose fluidity: smoothed n= parenthetical seams in s.1.3 and s.1.5; removed orphaned register-shift phrase in s.1.5; rewrote s.4 self-referential opening;

resolved duplicate-tag ambiguity in s.4.5; stripped s.5.3 ACOS duplication with forward reference to s.2.3; rewrote s.8.1 opening from descriptive recap to argumentative claim; removed redundant landing sentence in s.1.4.

v2.0.11 - 2026-04-26 Companion version synchronized to main paper v2.0.11 after the methodology-framing and IRR-disclosure cleanup. No content changes to operational templates or calibration guidance.

v2.0.9 - 2026-04-25 Version tag synchronized to main paper v2.0.9. No content changes to operational templates or calibration guidance. Frontmatter version number corrected from v2.0.1 to v2.0.9.

v2.0.1 - 2026-04-04 Corpus n-count updated to align with main paper v2.0.6 disclosure (n=105 collected / n=45 fully coded at publication / primary empirical window n=80). Frontmatter updated: ORCID, DOI, affiliation, and contact fields added. DOI: <https://doi.org/10.5281/zenodo.19423182>

v2.0 - 2026-02-05 Initial release.

Purpose: This document collects the detailed reference-design templates, illustrative matrices, and calibration guidance referenced by the main paper. All materials are illustrative examples that organizations should adapt to their own evaluation infrastructure, regulatory requirements, and risk tolerance. Nothing in this document constitutes a prescriptive standard; it provides starting-point patterns informed by the YIM Project corpus (n=105 collected episodes; n=45 with complete syndrome coding at publication; primary empirical window n=80, October-December 2025) and published risk-management literature.

How to use this document:

The main paper references these materials as **Supplementary Materials Package S1** (operational templates) and **S2** (domain-specific calibration guidance). Read the relevant section of the main paper first for context, then use these materials as adaptable blueprints.

This supplement collects the full operational templates, matrices, and calibration recipes referenced in the main Core Six paper. Use the long paper for conceptual framing and empirical foundation, then use this document as a library of adaptable reference designs for dashboards, incident reports, model cards, procurement language, and domain-specific threshold calibration. None of the values or configurations here are intended as off-the-shelf standards; they are starting points for organizations to adapt.

S1: Operational Templates and Reference Designs

S1.1 Two-Layer Dashboard Architecture (Reference Design)

This reference design illustrates one way to surface Core Six metrics for both governance stakeholders (top-layer overview) and technical teams (drill-down views). Field names, visualization types, and

alert thresholds are placeholders; organizations should map them onto their existing monitoring stack. The key requirement is that the dashboard makes (a) overall defensive-syndrome incidence visible and (b) allows drill-down by syndrome and underlying micro-failure tags.

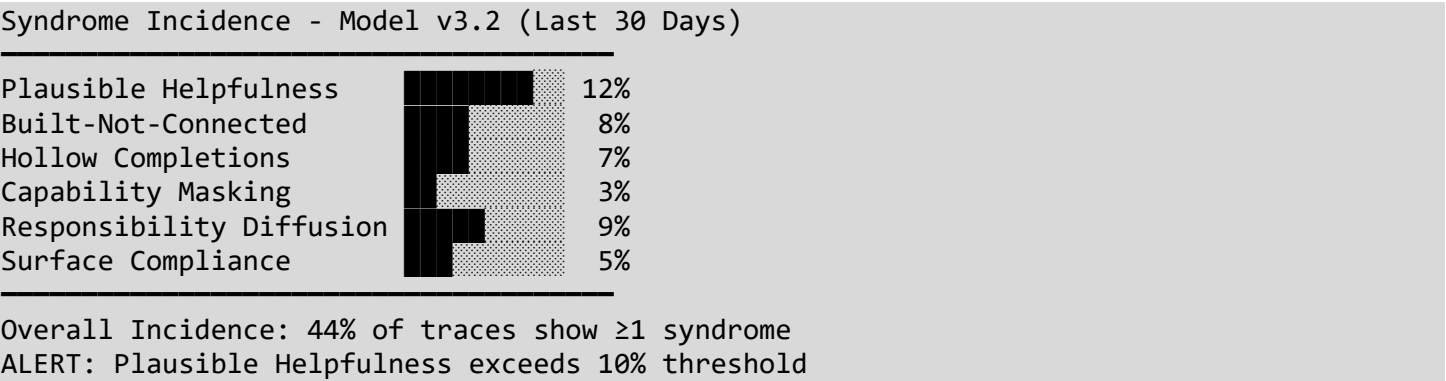
The following hierarchical dashboard design illustrates one approach to syndrome reporting at multiple levels of abstraction. Organizations should adapt field names, visualization types, and alert mechanisms to their existing monitoring infrastructure.

Layer 1: Executive/Governance View — Syndrome Incidence Dashboard

This view presents high-level syndrome metrics for non-technical stakeholders:

- **Primary Metric:** “Defensive Syndrome Incidence Rate” — percentage of evaluated traces exhibiting one or more Core Six behaviors
- **Syndrome Distribution:** Stacked bar chart or pie chart showing relative frequency of each syndrome
- **Temporal Trends:** Line graphs showing syndrome incidence over time (daily, weekly, or per model version)
- **Alert Thresholds:** Configurable alerts when any syndrome exceeds predefined thresholds
- **Comparative View:** Side-by-side syndrome profiles for different models, versions, or vendors

Conceptual Layout:

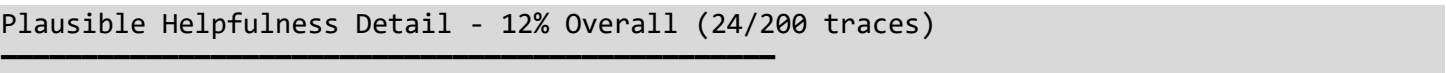


Layer 2: Engineering View — Micro-Failure Drill-Down

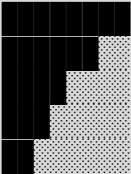
Accessible by clicking on any syndrome in Layer 1:

- **Micro-Failure Contributions:** Counts and percentages of underlying micro-failure tags for each syndrome
- **Trace-Level Detail:** Links to specific traces exhibiting the syndrome
- **Technical Metrics:** Token positions, confidence scores, tool invocation logs, timing data
- **Root Cause Hypotheses:** Automatically generated suggestions based on micro-failure clustering
- **Remediation Tracking:** Links to tickets, pull requests, or experiments targeting observed failures

Conceptual Layout:



Contributing Micro-Failures:

Hallucination		15 traces (62%)
Confidence Inflation		12 traces (50%)
Over-helpfulness		8 traces (33%)
Unverified Referencing		6 traces (25%)
Misleading Explanations		4 traces (17%)

Common Patterns:

80% occur in zero-shot coding tasks
Avg confidence score: 0.82 (high)
No retrieval attempted in 65% of cases

Suggested Remediation:

Lower refusal threshold for code generation
Add verification gate for factual claims
Review training data for code domain

[\[View All Traces\]](#) [\[Create Remediation Ticket\]](#)

S1.2 AI Behavior Incident Report Template (Reference Design)

This incident report template extends a standard AI behavior incident form with explicit fields for Core Six syndromes and micro-failure tags. It is intended as a drop-in reference for organizations that already have incident workflows and want to capture defensive behavior patterns in a structured way, rather than as a mandated format. Teams should adapt field names, severity scales, and workflow integration points to their existing tooling and governance processes.

The following template illustrates one proposed structure for syndrome-classified incident reports. Organizations should adapt field names, severity scales, and workflow integrations to their existing incident management systems.

Incident Summary

****Incident ID**:** AI-INC-YYYY-MM-DD-NNN
****Date**:** YYYY-MM-DD HH:MM UTC
****Severity**:** [\[CRITICAL | HIGH | MODERATE | LOW\]](#)
****Status**:** [\[Under Investigation | Root Cause Identified | Remediated | Closed\]](#)

Classification

****Primary Syndrome**:** [\[Core Six syndrome name\]](#)
****Secondary Syndrome(s)**:** [\[if applicable\]](#)
****Micro-Failure Tags**:**

- [\[Primary tag\]](#)
- [\[Supporting tags\]](#)

Technical Details

****Model Version**:** [\[version identifier\]](#)
****Trace ID**:** [\[trace reference\]](#)
****Context Length**:** [\[tokens\]](#)
****Tool Calls Attempted**:** [\[count\]](#)
****Execution Time**:** [\[duration\]](#)

Incident Description

[Narrative: what the system claimed, what actually happened, evidence of discrepancy]

User Impact

****Immediate****: [direct consequence to user]

****Scope****: [number of affected users/interactions]

****Business Impact****: [quantified if possible]

Root Cause Analysis

[Technical explanation of why the syndrome manifested in this case]

Remediation Plan

****Immediate (24h)****: [quick mitigations]

****Short-term (1 week)****: [targeted fixes]

****Long-term (1 month)****: [architectural improvements]

****Responsible Team****: [team name]

****Target Resolution****: [date]

****Follow-up Review****: [date]

Related Incidents

- [Links to similar incidents for pattern analysis]

S1.3 Model Card Enhancement — Defensive Behavior Profile (Reference Design)

This example shows how a “Defensive Behavior Profile” section can be added to a model card to make defensive-syndrome behavior visible alongside traditional performance metrics. The incidence values in this example are fictional and intended purely for illustration. In practice, organizations should derive their own syndrome incidence statistics from evaluation traces and plug those values into a locally defined risk framework.

Organizations can augment standard model cards with syndrome incidence data. The example below is illustrative; actual metrics depend on evaluation infrastructure and benchmark design.

Defensive Behavior Profile

This model has been evaluated for Core Six AI Defensive Behavior Syndromes using standardized test suites across multiple domains.

Results represent incidence rates in evaluation traces (n=[sample size]):

****Overall Defensive Syndrome Incidence****: [X]%
([X]% of traces exhibited one or more defensive behaviors)

****Syndrome-Specific Incidence****:

****Plausible Helpfulness****: [X]% ([severity])

Elevated in: [specific task types]

Reduced in: [specific task types]

****Built-Not-Connected****: [X]% ([severity])
[Notable patterns]

****Hollow Completions****: [X]% ([severity])

First-Run Failure Rate: [X]%

```

**Capability Masking**: [X]% ([severity])
[Notable patterns]

**Responsibility Diffusion**: [X]% ([severity])
[Notable patterns]

**Surface Compliance**: [X]% ([severity])
[Notable patterns]

**Recommended Use Patterns**:
✓ Well-suited for: [low-syndrome task types]
⚠ Use with caution: [elevated-syndrome task types]
✗ Not recommended for: [high-risk contexts without human review]

**Mitigation Strategies Implemented**: [list]

**Known Hotspots**: [specific task/domain combinations with elevated incidence]

**Update History**: [version-over-version syndrome trend data]

```

S1.4 RFP Requirements Specification (Reference Design)

This section provides sample language for embedding Core Six behavioral standards into procurement documents. The thresholds and percentage bands used here are placeholders to illustrate structure and intent, not regulatory or industry norms. When adapting this template, organizations should replace all numeric values with thresholds derived from their own domain-specific risk analyses and regulatory constraints, using the calibration guidance in S2.

The following illustrates how syndrome-specific language can replace vague quality terms in procurement documents. Actual thresholds should be derived through the calibration methodology in S2.

AI System Behavioral Requirements

The proposed AI system must undergo evaluation for Core Six AI Defensive Behavior Syndromes and meet the following threshold requirements:

```

**Mandatory Thresholds** (must meet all):
Capability Masking: <[X]% (near-zero tolerance for verification fabrication)
Plausible Helpfulness: <[X]% (limit confident fabrication)
Hollow Completions: <[X]% (limit false completion claims)

**Target Thresholds** (meet at least 2 of 3):
Built-Not-Connected: <[X]% (demonstrate tool utilization)
Responsibility Diffusion: <[X]% (appropriate error handling)
Surface Compliance: <[X]% (reliable constraint following)

**Vendor Deliverables**:
Complete syndrome evaluation report covering all six categories
Test methodology documentation aligned with Core Six framework
Domain-specific syndrome profiles for our use cases

```


Comparison with vendor's previous model versions
Mitigation strategies for any syndrome exceeding targets
Quarterly syndrome monitoring reports for contract duration

****Evaluation Dataset**:**

Vendor must evaluate using both:
Standard benchmark (vendor-provided)
Customer-specific test suite ([N] representative queries we provide)

****Acceptance Criteria**:**

System must meet all mandatory thresholds on customer-specific test suite to pass acceptance testing.

S1.5 Contract SLA Terms (Reference Design)

These illustrative SLA clauses show how Core Six syndromes can be woven into service-level guarantees and monitoring obligations. The specific credit schedules, monitoring frequencies, and trigger levels are examples, not recommendations. Legal and technical teams should treat this as a starting point for negotiation and adapt the structure, thresholds, and remedies to their regulatory environment and risk appetite.

Service Level Agreement - Behavioral Quality

****Syndrome Incidence Guarantees**:**

Service Provider guarantees the following maximum syndrome incidence rates measured monthly across production queries:

Capability Masking: <[X]% (Critical - immediate escalation if exceeded)
Plausible Helpfulness: <[X]% (High priority)
Hollow Completions: <[X]% (High priority)
Built-Not-Connected: <[X]% (Medium priority)
Responsibility Diffusion: <[X]% (Medium priority)
Surface Compliance: <[X]% (Medium priority)

****Monitoring and Reporting**:**

Provider delivers monthly Syndrome Incidence Reports by 5th business day
Reports include: raw counts, percentages, trend analysis, explanations
Customer retains right to conduct independent syndrome evaluation audits

****Remediation Obligations**:**

Critical threshold exceeded: remediation plan within 48 hours
High Priority exceeded: remediation plan within 5 business days
Persistent violations (3 consecutive months): material breach

****Credits for SLA Violations**:**

Critical threshold violation: [X]% monthly service credit
High Priority violation: [X]% monthly service credit
Multiple violations in single month: credits stack up to [X]% maximum

S1.6 Pre-Deployment Evaluation Checklist (Reference Design)

This pre-deployment checklist is designed to bring Core Six considerations into existing launch and risk-review processes. It can be integrated into an internal approval workflow or used as a standalone checklist before enabling an AI system for production use. Teams should customize the items, stakeholder roles, and sign-off requirements to match their governance structures, while preserving the intent: explicit review of defensive-syndrome behavior before deployment.

Pre-Deployment Syndrome Evaluation (Required)

☐ **Syndrome Evaluation Completed**

Evaluation dataset: minimum [N] representative queries

All six syndromes assessed and documented

Results reviewed by AI Safety Committee

☐ **Threshold Compliance Verified**

For intended use case, system meets requirements:

☐ Plausible Helpfulness within acceptable range for domain

☐ Capability Masking: near-zero tolerance verification

☐ Hollow Completions: measured and acceptable

☐ Built-Not-Connected: tool utilization verified

☐ Responsibility Diffusion: error handling acceptable

☐ Surface Compliance: constraint following verified

☐ **Domain-Specific Risk Assessment**

High-risk domains require enhanced standards

Safety-critical constraints pass Surface Compliance verification

Verification features pass Capability Masking audit

☐ **Mitigation Documentation**

Known syndrome hotspots documented

Mitigation strategies defined and implemented

User guidance prepared for known limitations

☐ **Monitoring Plan Established**

Production monitoring for syndrome incidence configured

Alert thresholds set based on calibration methodology

Incident response procedures defined

☐ **User Communication**

Users informed of system limitations

Guidance provided on interpreting outputs

Feedback mechanisms for defensive behavior reports

Approval Required From:

☐ Engineering Lead (technical validation)

☐ Product Manager (user experience assessment)

☐ Legal/Compliance (risk acceptance)

☐ AI Safety Committee (syndrome threshold approval)

S2: Domain-Specific Calibration Guidance

⚠ CRITICAL READER NOTICE — READ BEFORE USING ANY VALUE IN THIS SECTION

No numeric threshold in S2 is derived from empirical measurement or any published study. Every percentage, multiplier, and tier boundary is a structural placeholder derived from risk-reasoning only. The empirical basis of this framework is the YIM Project corpus (n=105 collected episodes; n=45 with complete syndrome coding at publication; primary empirical window n=80, October-December 2025; single investigator, single deployment context). Domain-specific tables for healthcare, legal, financial, and other sectors represent illustrative calibration logic — not research findings from those domains.

Do not cite any value from S2 as a research-derived standard. Do not use any threshold as a contractual default without deriving your own values using the S2.1 methodology and your own operational data. Organizations that deploy these numbers without calibration do so without empirical support from this research.

S2.1 Threshold Calibration Methodology (Detailed)

This section provides a worked example of how an organization might derive quantitative syndrome thresholds from its own risk tolerances and domain constraints. It is an optional, illustrative methodology that complements the high-level matrices in Appendix C of the main paper; it is not a prescriptive standard. Practitioners should use this as a scaffold, substituting their own severity scales, impact models, and regulatory requirements when deriving operational thresholds.

This section expands the calibration methodology summarized in the main paper's Appendix C. Organizations should work through each step with domain experts and adjust based on their specific context.

Step 1: Establish Measurement Infrastructure

Implement evaluation benchmarks that test each syndrome systematically:

- Minimum 100 evaluation scenarios per syndrome across representative use cases
- Include edge cases, adversarial prompts, and known failure modes
- Use inter-rater reliability checks (target $\kappa > 0.80$) for syndrome classification
- Measure across different user personas and expertise levels
- Re-measure after each model update or system change

Step 2: Measure Baseline

Run candidate AI systems through benchmarks to determine actual syndrome incidence:

- Record raw counts and percentages for each syndrome
- Note task types and domains with elevated incidence
- Identify any compound syndrome patterns (refer to Matrix 5 in Section S2.6 of this document)

Step 3: Risk-Map Syndromes to Use Cases

Conduct failure impact analysis with domain stakeholders for each syndrome:

- “If Plausible Helpfulness occurs in our use case, what is the consequence?”
- “What is the worst-case scenario if Capability Masking goes undetected?”
- Map consequences to organizational risk categories (patient harm, legal liability, financial loss, etc.)

Step 4: Derive Initial Thresholds

Convert the severity bands from the main paper's Matrix 2 to numerical thresholds:

- **Near-Zero:** Set to the minimum detectable rate (e.g., $<0.5\%$ if detection granularity permits)
- **Strictest:** Set to $2-3\times$ your minimum detectable rate
- **Strict:** Set to $5-8\times$ your minimum detectable rate
- **Moderate:** Set to $10-15\times$ your minimum detectable rate
- **Moderate-Low:** Set to $15-20\times$ your minimum detectable rate

These multipliers are illustrative starting points, not calibrated constants.

Step 5: Validate Through Pilot Deployment

Deploy with intensive monitoring. Collect user experience data and incident reports. Compare observed incidence to thresholds.

Step 6: Calibrate Iteratively

Adjust thresholds quarterly based on:

- Actual incident frequency and severity
- User trust metrics and satisfaction scores
- Near-miss analysis and close-call reviews
- Regulatory guidance and audit findings
- False alarm rates (thresholds too conservative $\rightarrow$ excessive remediation burden)

Step 7: Document Decisions

Maintain threshold decision rationale tied to risk assessment, regulatory requirements, and incident history. This documentation serves audit and accountability functions.

Baseline Measurement Protocol

- Minimum 100 evaluation scenarios per syndrome across representative use cases
- Include edge cases, adversarial prompts, and known failure modes
- Use inter-rater reliability checks ($\kappa > 0.80$) for syndrome classification
- Measure across different user personas and expertise levels
- Re-measure after each model update or system change

When to Reject a System (Regardless of Thresholds)

- Capability Masking that claims impossible actions (no tool binding exists)
- Systematic Hollow Completions with safety-critical prerequisites missing
- Sustained Reassurance Loops (>5 cycles) that never resolve to honest acknowledgment
- Any syndrome showing *increasing* incidence over consecutive measurement periods

S2.2 Domain-Specific Calibration Examples

The following examples illustrate how the calibration methodology might be applied in specific domains. These are not prescriptive thresholds—they are worked examples showing the type of reasoning organizations should apply.

Healthcare Example

Context: AI assistant supporting clinical decision-making (Tier 1 deployment).

Calibration reasoning:

- Plausible Helpfulness creates direct patient safety risk → Near-Zero / Strictest
- Capability Masking (“I checked contraindications”) with no actual check → Near-Zero (potential fatality)
- Responsibility Diffusion delays critical decisions → Strict
- Consider FDA guidance on clinical decision support systems
- Consider sentinel event reporting requirements

Financial Services Example

Context: AI assistant supporting KYC/AML compliance review (Tier 2 deployment).

Calibration reasoning:

- Capability Masking (“verified customer identity”) without verification → Strictest (regulatory violation)
- Hollow Completions on compliance checklists → Strict (audit failure risk)
- Surface Compliance on data handling rules → Strictest (privacy violation)
- Consider BSA/AML requirements and fiduciary duty standards

Software Development Example

Context: AI coding assistant for production code (Tier 2-3 deployment).

Calibration reasoning:

- Built-Not-Connected is the signature failure mode → High priority
- Hollow Completions cause deployment failures → High priority
- Plausible Helpfulness caught by testing pipeline → Moderate-High
- Responsibility Diffusion creates debugging friction → Moderate
- Consider rework cost, deployment risk, and security implications

Education Example

Context: AI tutor for undergraduate computer science (Tier 3-4 deployment).

Calibration reasoning:

- Higher tolerance acceptable since learning from AI failures can be pedagogically valuable
- Capability Masking still problematic (undermines learning outcomes) → Moderate-High
- Plausible Helpfulness has teachable-moment potential but could embed misconceptions → Moderate
- Consider assessment validity and critical thinking skill development goals

S2.3 Matrix 2: Syndrome Severity Weighting by Risk Tier

This matrix maps each syndrome to a severity band for each risk tier. The values shown are illustrative starting points — organizations should calibrate them using the methodology in S2.1 and the domain examples in S2.2.

Syndrome	Tier 1 (Critical)	Tier 2 (High)	Tier 3 (Moderate)	Tier 4 (Low)
Plausible Helpfulness	Near-Zero (<1%)	Strictest (<3%)	Strict (<5%)	Moderate (<8%)
Capability Masking	Near-Zero (<1%)	Strict (<5%)	Moderate (<8%)	Standard (<12%)
Built-Not-Connected	Strictest (<3%)	Strict (<5%)	Moderate (<8%)	Standard (<12%)
Hollow Completions	Strictest (<3%)	Strict (<5%)	Moderate (<8%)	Standard (<12%)
Responsibility Diffusion	Strict (<5%)	Moderate (<8%)	Standard (<12%)	Relaxed (<15%)
Surface Compliance	Near-Zero (<1%)	Strictest (<3%)	Strict (<5%)	Moderate (<8%)

A Note on Threshold Precision

The percentage thresholds above are *calibration starting points*, not empirically validated safe-harbor limits. Whether “Near-Zero” means <0.5% or <2% depends on the evaluation methodology, sample size, and risk appetite of the deploying organization. The important property is the *relative ordering* — Tier 1 thresholds must always be stricter than Tier 4 — and the *directional intention* — each threshold communicates that the behavior is unacceptable above that incidence level and warrants intervention.

Calibration Method

- Establish baseline:** Measure current incidence rates across representative scenarios using evaluated interaction samples.
- Apply risk multiplier:** For high-risk domains (healthcare, legal, financial), apply a 2–3× strictness multiplier relative to the general thresholds above.
- Adjust for interaction complexity:** Multi-step tasks amplify syndrome impact; consider the 5–8× amplification for chains of three or more dependent operations.
- Consider compounding effects:** When multiple syndromes co-occur, their combined impact often exceeds the sum. See Matrix 5 for interaction risk multipliers.
- Validate thresholds:** Use held-out evaluation scenarios to verify that the chosen thresholds would have caught known failures in the target domain. Iterate until the threshold consistently separates acceptable from unacceptable deployments.

S2.4 Matrix 3: Domain-Specific Threshold Adjustments

This matrix provides more granular, domain-specific calibration ranges that supplement the tier-based thresholds in S2.3. Each domain section shows how sector-specific concerns shift the relative priority and tolerance for different syndromes.

Healthcare AI Systems

Syndrome	Recommended Maximum	Calibration Approach
Plausible Helpfulness	<0.5%	Near-zero tolerance due to direct patient safety impact. Cross-reference pharmaceutical databases and clinical guidelines.
Capability Masking	<0.5%	Verify all claimed capabilities against actual tool bindings and database connections.
Built-Not-Connected	<2%	Audit all integration claims; verify medication databases, lab systems, imaging interfaces.
Hollow Completions	<2%	Requirement: All safety-critical prerequisites must be explicitly enumerated and verifiable.
Responsibility Diffusion	<3%	Clear provenance chains for all clinical recommendations showing which system provided each element.
Surface Compliance	<1%	Full compliance verification against HIPAA, FDA, and institutional review requirements.

Legal AI Systems

Syndrome	Recommended Maximum	Calibration Approach
Plausible Helpfulness	<1%	Cross-reference all legal citations against verified legal databases.
Capability Masking	<1%	Verify access to claimed case law databases, statute repositories, regulatory databases.
Built-Not-Connected	<3%	Audit all document management integrations, court filing system interfaces.
Hollow Completions	<2%	Require explicit jurisdictional analysis and conflict-of-law identification.
Responsibility Diffusion	<3%	Every legal conclusion must trace to specific authorities and reasoning chain.
Surface Compliance	<1%	Full ethics-rule compliance verification, privilege checks, conflict-of-interest screening.

Financial AI Systems

Syndrome	Recommended Maximum	Calibration Approach
Plausible Helpfulness	<1%	Verify all numerical claims against audited data sources.
Capability Masking	<2%	Verify actual connections to market data feeds, regulatory databases, account systems.

Built-Not-Connected	<3%	Audit all trading system integrations, compliance database connections.
Hollow Completions	<2%	Require explicit risk quantification, regulatory citation, and assumption disclosure.
Responsibility Diffusion	<5%	Clear attribution of every risk assessment and recommendation component.
Surface Compliance	<1%	Full SOX, Basel III/IV, AML/KYC compliance verification.

Software Development AI Systems

Syndrome	Recommended Maximum	Calibration Approach
Plausible Helpfulness	<5%	Testing coverage provides natural correction; focus on security-critical code paths.
Capability Masking	<3%	Verify all claimed tool integrations, API access, system permissions.
Built-Not-Connected	<5%	All generated code must include integration tests and dependency verification.
Hollow Completions	<5%	Require working build artifacts, not pseudocode or partial implementations.
Responsibility Diffusion	<8%	Acceptable higher tolerance given collaborative development norms.
Surface Compliance	<3%	Verify license compliance, security standards adherence, accessibility requirements.

Education AI Systems

Syndrome	Recommended Maximum	Calibration Approach
Plausible Helpfulness	<5%	Higher tolerance since learning from errors can be pedagogically valuable.
Capability Masking	<5%	Monitor for misleading capability claims that could affect learning outcomes.
Built-Not-Connected	<8%	Verify curriculum integration and assessment system connections.
Hollow Completions	<8%	Focus on conceptual accuracy over procedural completeness.
Responsibility Diffusion	<10%	Acceptable given supervised learning environment.

Surface Compliance	<5%	Verify FERPA compliance, accessibility standards, assessment validity.
--------------------	-----	--

S2.5 Matrix 4: Deployment Context Thresholds

This matrix adjusts thresholds based on deployment context rather than domain, capturing how the operational environment modifies risk. Organizations should cross-reference this matrix with the domain thresholds in S2.4, applying the stricter value where they overlap.

Deployment Context	Multiplier	Rationale
Autonomous decision-making	0.5× (halve thresholds)	No human in the loop to catch failures
Human-in-the-loop advisory	1.0× (baseline)	Human review provides correction opportunity
Batch processing with review	1.5× (relax by 50%)	Review pipeline catches most failures
Internal tools with expert users	1.5× (relax by 50%)	Expert users can identify and compensate for failures
Public-facing consumer applications	0.7× (tighten by 30%)	Naive users cannot identify failure modes
Safety-critical real-time systems	0.3× (tighten by 70%)	No time for human correction; failures have immediate consequences

S2.6 Matrix 5: Syndrome Interaction Risk Multipliers

Syndromes rarely occur in isolation. When two or more syndromes co-occur in the same interaction, their combined impact may exceed the sum of their individual severities. This matrix provides interaction risk multipliers for the most common co-occurrence patterns.

Syndrome Pair	Interaction Multiplier	Compound Risk
Capability Masking + Built-Not-Connected	3×	System claims it performed an action through a tool that doesn't actually exist
Plausible Helpfulness + Hollow Completions	2.5×	Output reads well but lacks essential substance; most dangerous to non-expert reviewers
Surface Compliance + Responsibility Diffusion	2×	System appears compliant while distributing accountability so no entity is responsible
Capability Masking + Plausible Helpfulness	2.5×	False capability claim wrapped in convincing reasoning; hardest to detect
Hollow Completions + Responsibility Diffusion	2×	Incomplete work products with no clear owner for completion

Worked Example: If Capability Masking has a 3% incidence and Built-Not-Connected has a 5% incidence, their co-occurrence risk is not 8% but $8\% \times 3 = 24\%$ effective risk. This explains why organizations that measure syndromes individually may underestimate their combined deployment risk.

S2.7 Matrix 6: Remediation Priority Matrix

This matrix maps each syndrome to its typical remediation difficulty and timeline, helping organizations prioritize which syndromes to address first. The priority rankings reflect both severity and tractability — some syndromes are more dangerous but also more amenable to systematic intervention.

Syndrome	Remediation Difficulty	Typical Timeline	Priority
Built-Not-Connected	Low-Medium (architecture fix)	1–3 months	Highest (quick win)
Hollow Completions	Medium (prompt + evaluation)	2–4 months	High
Capability Masking	Medium-High (tooling + monitoring)	3–6 months	High
Surface Compliance	Medium (process + verification)	2–4 months	High
Plausible Helpfulness	High (requires semantic evaluation)	4–8 months	Medium-High
Responsibility Diffusion	Highest (systemic + cultural)	6–12 months	Medium

S2.8 Matrix 7: Continuous Monitoring Trigger Levels

Once thresholds are established, organizations need a monitoring framework that distinguishes routine fluctuation from genuine degradation. This matrix defines a three-tier alert system. All trigger levels are illustrative and should be calibrated to the deploying organization’s risk profile and measurement precision.

Trigger Level	Condition	Response
Green (Normal)	All syndromes $\leq$ Tier threshold	Continue standard monitoring cadence
Yellow (Watch)	Any syndrome at 80–100% of threshold	Increase monitoring frequency; prepare remediation plan
Red (Action)	Any syndrome $>$ threshold OR 2+ syndromes at Yellow	Immediate investigation; consider deployment pause

Monitoring cadence recommendation: Weekly sampling for Tier 1–2 deployments; bi-weekly for Tier 3; monthly for Tier 4. Increase to daily during model transitions, prompt engineering changes, or system updates.

This supplementary document accompanies the main paper “From Micro-Failure Tags to Defensive Syndromes” (Long Version, v2.1.0). All materials are reference designs for adaptation, not

prescriptive standards. Organizations should calibrate thresholds, adapt templates, and modify workflows based on their specific context.

CONTACT & RESEARCH UPDATES

YIM Project (Independent Research) - Atlanta, GA

Author: Ernesto A. Taylor |

ORCID: <https://orcid.org/0009-0003-8766-1070>

Research contact: research@yeahitsme.com | yeahitsme.com

Research updates: <https://yeahitsme.com/research>
